

Predicting the Equity Premium with Dividend Ratios: Reconciling the Evidence

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Abstract

This paper evaluates the ability of dividend ratios to predict the equity premium. We conduct an out-of-sample comparative study and apply the Goyal and Welch (2003) methodology to equity premia derived from the UK FTSE All-Share and the S&P 500 indices. Preliminary in-sample univariate regressions reveal that in both markets the equity premium contains an element of predictability. However, out-of-sample the considered models outperform the historical moving average only in the UK context. This is confirmed by the graphical diagnostic which further indicates that dividend ratios are useful predictors of UK excess returns. Our paper provides a possible explanation of why dividend ratios might be more informative in the UK market and links these findings to a phenomenon that is not widely known in the predictability literature. Finally, Campbell and Shiller (1988b) identities are employed to account for the time-varying properties of the dividend ratio and dividend growth processes. It is shown that by instrumenting the models with the identities, forecasting ability can be further improved.

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1. Introduction

The predictability of stock market returns is one of the most controversial and intensely debated issues in empirical finance. A voluminous literature around the issue has evolved during the last two decades, rendering its overall assessment extremely difficult. The dividend-to-price ratio and the dividend yield rank amongst the most popular candidates that can be used to predict stock market returns (Rozeff, 1984; Campbell and Shiller, 1988b; Fama and French, 1988; Hodrick, 1992; Lewellen, 2004).¹ However, finding statistical significance based on in-sample tests is not conclusive evidence that these variables have out-of-sample power or that predictability is also economically significant. Bossaerts and Hillion (1999), Goyal and Welch (2003) and Ang and Bekaert (2007) cast doubt on the evidence documented by the early authors and report negative results when they use out-of-sample tests.

In addition, the fact that the existing literature relies mainly on in-sample tests raises concerns of data mining. Interestingly, although it is typically believed that out-of-sample tests are able to correct for this and to provide a measure against data mining, this “conventional wisdom” has been challenged by a few researchers. For example, Inoue and Killian (2004) and Rapach and Wohar (2006) both show that, if appropriate tests are employed, in-sample and out-of-sample tests are equally reliable.²

¹ Other financial and macroeconomic predictive variables include the earnings-to-price ratio (Campbell and Shiller, 1988a; Campbell and Shiller, 1998; Lamont, 1998), the book-to-market ratio (Kothari and Shanken, 1997; Pontiff and Schall, 1998), the short-term interest rate (Hodrick, 1992; Ang and Bekaert, 2007), the yield spread (Keim and Stambaugh, 1986; Campbell, 1987; Fama and French, 1989), and more recently the consumption-wealth ratio (Lettau and Ludvigson, 2001).

² Rapach and Wohar (2006) argue that the case for a predictable component in stock returns is strengthened by this result.

This paper provides a comparative study and evaluates the out-of-sample performance of dividend ratios when used to predict the equity premium across the US and UK stock markets. Our study adds to an emerging literature that is linked to the UK market, in contrast to the overwhelming majority of the extant literature which focuses primarily on the US market. Recent studies (e.g., Pesaran and Timmermann, 2000; and McMillan, 2003) suggest that dividend yields may have some predictive power in the UK context while evidence from the US market reveals the poor out-of-sample performance of the dividend ratios (e.g., Goyal and Welch, 2003; and Ang and Bekaert, 2007).

Our paper seeks to shed some light on the matter and offers a possible explanation of the contradictory findings between the two markets. In particular, we consider the study of Fama and French (2001) who present the phenomenon of the disappearing dividend in the US and document the tendency of publicly traded non-financial non-utility firms to pay less dividends since 1978. More recently, Amihud and Li (2006) also propose an explanation for the “disappearing dividend” phenomenon in the US market which they argue that is partly due to a decline in the information content of dividend announcements.

To our knowledge, extremely limited work has been documented in the UK context. Only Benito and Young (2003) examine the issue and uncover an increasing proportion of UK quoted companies omitting cash dividends. Considering the above, we obtained annual dividends data for both markets and we provide further empirical evidence on the proportions of companies that actually issue dividends for the period 1975-2004. We argue that although in both markets, there is a tendency of firms to pay less dividends, the phenomenon is not so pronounced in the UK. Therefore it is

possible that dividends conveyed more information during the out-of-sample period under consideration in our study.

Turning to our approach, we adopt the Goyal and Welch (2003) (henceforth GW) methodology which will allow us to re-examine the predictive ability of the dividend ratios and observe dynamically when predictability has occurred (or not) in the two markets. Our empirical application relies on the FTSE All Share index which is the most commonly used index with respect to stock return predictability in the UK market. The US data are derived from the S&P 500 index. The data set constructed contains monthly observations covering the period from 1975 to 2004.

GW suggest a simple, recursive residuals (out-of-sample) graphical approach to assess equity premium prediction. An attractive attribute of this methodology is that it allows the identification of different periods of predictability. In other words, the out-of-sample performance of the dividend ratios can be decomposed, allowing detection of which month or year they succeed (or fail) in predicting the equity premium. The graphical diagnostic employed makes it easy to observe the relative performance of different forecasting models. The methodology is straightforward to implement and provides a powerful complement to those methods seeking to compute the appropriate standard errors of a test statistic. GW examine the comparative performance of the dividend yields and dividend-price ratios against the unconditional equity mean of the US equity premium. Their graphical diagnostic, employing annual data and covering the period from 1926 to 2002, indicates that the presumed predictive ability of the dividend ratios is a “mirage”, any predictive ability is dependent on only two years, 1973 and 1974. Strikingly we find strong support for the predictability of the UK equity premium. This is found using the graphical diagnostic and conventional in-sample and out-of-sample tests. In particular the graphical

diagnostic highlights the period from the late 1990s onward as highly predictable. On the other hand, when the graphical method is applied to our monthly US data, we find that indeed there is not much evidence in favour of out-of-sample predictability.

Finally, with the purpose of delving deeper into our main predictive variable, the dividend ratio, we employ the Campbell and Shiller (1988b) theory to account for the time-varying properties of the dividend yield and dividend growth processes. It is shown that by instrumenting the models with the Campbell and Shiller identities, forecasting ability can be yet further improved within the UK context.

The paper is organized as follows: Section 2 provides information on the data set and describes the main variables as well as the transformations used in the empirical analysis. Section 3 presents the methodology while section 4 offers the results and discusses the empirical findings. A final section concludes our study.

2. Data

The FTSE All Share, or to give its full name, “The Financial Times Actuaries All Share Index” is the most comprehensive UK stock market index. In our study, monthly data is employed on this index, covering the period from 1975:04 to 2004:10. The corresponding data for the US market were derived from the S&P 500 composite index. All data were obtained from Datastream, the starting date representing the extent of UK data availability from that databank. The use of a monthly frequency is particularly advantageous, allowing the examination of intra-annual predictability.

A number of variables are relevant to the study. Firstly, and as is common in the literature, log returns on the index are used:

$$r_{m,t} = \log[R_{m,t}] = \log[(P_t + D_t) / P_{t-1}] \quad (1)$$

where P is the stock index level and D the paid dividends. Next, the log returns on the three-month risk-free Treasury bill (called $R_{f,t}$) are calculated:

$$r_{f,t} = \log[1 + R_{f,t}] \quad (2)$$

The dividend-price ratio DP_t is the log of the aggregate dividends D_t divided by the aggregate stock market value P_t :

$$DP_t = \log[D_t / P_t] \quad (3)$$

whilst the dividend yield ratio DY_t is defined:

$$DY_t = \log[D_t / P_{t-1}] \quad (4)$$

Finally, the equity premium is denoted by EQP_t and is the return on the stock market ($r_{m,t}$) minus the return on a short-term risk-free treasury bill ($r_{f,t}$):

$$EQP_t = r_{m,t} - r_{f,t} \quad (5)$$

Table 1 provides the descriptive statistics for the series. For the entire sample period, the average UK log equity premium was 5.29%; the average log dividend-price ratio was -3.20 and the average log dividend yield was -3.19 (the respective values of the dividend ratios were 4.28% and 4.31%). Regarding the US market, the average log equity premium was 4.25%; the average log dividend-price ratio was -3.53 and the average log dividend yield was -3.51 (their respective values were 3.28% and 3.29%). In common with the vast majority of previous work, the augmented Dickey and Fuller (1979) test suggests the presence of a unit root in the dividend ratios. For completeness, Figure 1 plots the time series of the equity premium and the dividend ratios.

[Insert Table 1 around here]

[Insert Fig. 1 around here]

3. Methodology

3.1 In-Sample Predictability

In order to evaluate the in-sample predictive ability, we estimate the following regression model:

$$EQP_t = \alpha + \beta x_{t-1} + \varepsilon_t \quad (6)$$

where the predictive variable x_{t-1} can be either the lagged dividend-price ratio (DP_{t-1}) or the lagged dividend yield (DY_{t-1}). The predictive ability of x_{t-1} is assessed by examining the t-statistic corresponding to $\hat{\beta}$, the OLS estimate of β in equation (6), as well as the goodness of fit measure, R^2 . The null hypothesis tested is that of no predictability, i.e. $\beta = 0$ against the alternative that there is predictability, i.e. $\beta \neq 0$.

3.2 Out-of-Sample Predictability

A market timing investor would be interested in knowing if s/he could take advantage of the dividend ratios in order to predict the equity premium. Thus, the question is how the “conditional dividend-ratio models” would perform when compared to the “unconditional historical equity premium model” (the prevailing simple moving average). As in GW, forecasting regressions are estimated only with then-available data. Both the conditional models and our naïve benchmark model are estimated as recursive forecasts to predict one-month-ahead equity premia.

Our next goal is to compare the out-of-sample forecasts from the dividend model predictive regressions against the historical mean. If the conditional dividend ratio model outperforms the prevailing moving average then this implies that the dividend ratios add useful information and improve predictive ability. First we report the statistics on the out-of-sample prediction errors obtained in different sample

periods. In particular we document the mean, standard deviation, root mean square error (RMSE) and mean absolute error (MAE) of equity premium prediction errors. Finally, we compute the Diebold and Mariano (1995) test statistic for equal predictive accuracy.

3.3 A Graphical Evaluation Method for the Out-of-Sample Performance

Following GW, we employ a graphical method as a complementary diagnostic for equity premium and stock return prediction. The procedure consists of plotting the cumulative sum-squared error from the unconditional model minus the cumulative sum-squared error from the dividend ratio model (denoted by $Net - SSE_T$ for a sample size of T observations). Expressed algebraically:

$$Net - SSE_T = \sum_{t=1}^T [SE_t^{Prevailing\ Mean} - SE_t^{Dividend\ Model}] \quad (7)$$

where SE_t is the squared out-of-sample prediction error in observation t . The unconditional SE is obtained when the prevailing up-to-date equity premium average is used to forecast the following month's equity premium. The conditional prediction errors of the dividend models are obtained from recursive regressions with either DP_{t-1} or DY_{t-1} as the sole predictor of the following month's equity premium. Clearly, a positive value indicates that the dividend ratio model has outperformed the unconditional model so far. In addition, a positive slope indicates that the dividend model had lower forecasting error in a given month (i.e. superior performance).

Although graphing recursive residuals is relatively simple, Goyal and Welch (2003) stress its neglect by the literature implies some possible insights regarding predictability have typically been overlooked. In particular, the methodology allows for a dynamic identification of predictability. Time periods where dividend ratios

succeed (or fail) in predicting the equity premium relative to the prevailing mean can be clearly observed. As a consequence the graphical procedure can be used to enhance information derived from more conventional summary measures. As noted in the introduction, GW claim that the graphical diagnostic reveals any predictability shown by US summary measures is caused primarily by outliers. On the other hand, it is possible that when the summary measures indicate no predictability, the graphical procedure may reveal pockets of forecastability which are hidden when an averaging procedure is employed.

3.4 Instrumenting the Changing Dividend-Ratio Process

GW argue that changes in dividend ratio autocorrelation and in the ability to predict changes in dividend growth could themselves imply changes in the dividend ratio's ability to predict the equity premium. These process changes can be used to enhance the dividend ratio forecasting coefficients for the equity premium. To explain this in more detail consider the Campbell and Shiller (1988b) approximate present value relation with time-varying expected returns. Assuming that dividends and returns follow log-linear processes, the approximation begins with the following identity:

$$1 \equiv (1 + R_{m,t+1})^{-1} (1 + R_{m,t+1}) \Rightarrow$$

$$1 \equiv (1 + R_{m,t+1})^{-1} \left(1 + \frac{P_{t+1} + D_{t+1} - P_t}{P_t}\right) \equiv (1 + R_{m,t+1})^{-1} \frac{P_{t+1} + D_{t+1}}{P_t} \quad (8)$$

After some algebra, we find that the log dividend-price ratio can be approximated by the following relationship:³

$$d_t - p_t \approx r_{m,t+1} - \Delta d_{t+1} - k + \rho(d_{t+1} - p_{t+1}) \quad (9)$$

or if we isolate returns on the LHS:

$$r_{m,t+1} \approx (d_t - p_t) - \rho(d_{t+1} - p_{t+1}) + \Delta d_{t+1} + k \quad (10)$$

³ Lower case letters denote logs while k and ρ are parameters of linearization.

Now taking covariances with DP_t and dividing by the variance of DP_t we have:

$$\begin{aligned} \frac{Cov(r_{m,t+1}, DP_t)}{Var(DP_t)} &\approx 1 - \rho \frac{Cov(DP_{t+1}, DP_t)}{Var(DP_t)} + \frac{Cov(\Delta d_{t+1}, DP_t)}{Var(DP_t)} \\ \Rightarrow \beta_{r_{m,t+1}, DP_t} &\approx 1 - \rho \beta_{DP_{t+1}, DP_t} + \beta_{\Delta d_{t+1}, DP_t} \end{aligned} \quad (11)$$

where $\beta_{Y,X}$ denotes the coefficient of X in a regression of Y on X and a constant.

The new model employs equation (11).⁴ To estimate (11) we initially carry out recursive estimations of the following regressions of the dividend-price ratio and the dividend growth on the lagged dividend-price ratio:

$$DP_{t+1} = \alpha_{0s} + \alpha_{1s} DP_t + u_{1t+1}, \quad t = 1, \dots, p+s-1, \quad s = 1, \dots, T-p \quad (12)$$

$$\Delta d_{t+1} = \gamma_{0s} + \gamma_{1s} DP_t + u_{2t+1}, \quad t = 1, \dots, p+s-1, \quad s = 1, \dots, T-p \quad (13)$$

where T denotes the sample size, p denotes the in-sample size and s is the number of out-of-sample observations. Once we have obtained the estimated recursive coefficients $\hat{\alpha}_{1s}$ and $\hat{\gamma}_{1s}$ from the above regressions, we can use (11) to calculate the instrumented betas $\tilde{\beta}_{1s}$ as follows:

$$\tilde{\beta}_{1s} = 1 - \rho \hat{\alpha}_{1s} + \hat{\gamma}_{1s}, \quad s = 1, \dots, T-p \quad (14)$$

Accordingly, using the above instrumented betas, the indirect Campbell and Shiller forecasts are constructed as follows:

$$\hat{r}_{m,t+1} = \hat{\beta}_{0s} + \tilde{\beta}_{1s} DP_t, \quad s = 1, \dots, T-p \quad (15)$$

On the other hand, direct forecasts (the straight dividend model) are constructed simply by using the equation:

$$r_{m,t+1} = \beta_{0s} + \beta_{1s} DP_t + u_{t+1}, \quad t = 1, \dots, p+s-1, \quad s = 1, \dots, T-p \quad (16)$$

and estimating recursively using OLS.

⁴ Note that these approximations work only for returns and not for equity premia and also for dividend-price ratios but not dividend yields.

4. Results

4.1 In-Sample Fit

Table 2 tabulates the results of the univariate regressions that associate the log equity premium with the lagged dividend-price ratio (DP_t) and the lagged dividend yield (DY_t). In order to give a more complete view of the in-sample performance of the dividend ratios, we also report results for two other periods which include observations either up to 1995 or to 2000.

[Insert Table 2 around here]

Looking at the full sample, both the dividend-price ratio and the dividend yield retain good statistical significance in the UK (the t-statistics are 3.76 and 4.08 respectively). As shown in Table 2, the produced t-statistics, both regular and Newey-West adjusted for heteroskedasticity and autocorrelation, vary between 1.90 and 4.08. Such results are not untypical of those found in the return predictability literature.

With respect to the US market, we also find in-sample statistical significance and the t-statistics for the dividend ratios, both regular and Newey-West adjusted for heteroskedasticity and autocorrelation, range between 2.00 and 2.89.

Overall, it can be said that our preliminary results show in-sample predictability for both markets which is congruent with much of the recent literature. However, it can also be said that when we use the full sample and we have maximum degrees of freedom for both dividend models, the UK regressions produce higher t-statistics (both normal and adjusted) and higher R-squareds compared to the corresponding US regressions. Hence, a higher degree of in-sample predictability seems to exist in the UK market.

4.2 Out-of-Sample Forecasts

Obviously, a trader or other interested party would not be able to use the in-sample results to forecast the equity premium. As such, Table 3 displays statistics on the out-of-sample prediction errors when only then-available data are used to construct forecasts. The errors from the dividend models are obtained from recursive regressions that employ the lagged dividend-price ratio and the lagged dividend yield as predictors for the equity premium. For a more thorough examination, we arbitrarily split the out-of-sample period into three sub-sample periods, each spanning approximately 5 years.

[Insert Table 3 around here]

Strikingly, both the dividend-price ratio and the dividend yield appear to outperform the historical moving average across all periods in the UK. This is particularly noticeable during the 2000-2004 sample period where the RMSE for our naive prevailing mean model is 6.10% whilst for the dividend-price ratio it is 5.35% and for the dividend yield it is 5.34%. Regarding the US market, the dividend ratios seem to perform slightly better during the full out-of-sample period and also during two out of the three chosen subsample periods. Clearly, we need to examine whether these identified differences are statistically significant. Table 4 reports the computed Diebold and Mariano (1995) (henceforth DM) statistics for different out-of-sample periods.

[Insert Table 4 around here]

In relation to the UK, for the full sample the dividend ratio models significantly outperform the historical moving average at the 5% level of significance (with the DM statistics being 2.13 for the dividend-price model and 1.70 for the dividend yield model). These results add to the small body of work suggesting, that perhaps in

contrast to the US, a degree of predictability exists in UK stock returns (see, *inter alia*, McMillan, 2003). Furthermore, although an arbitrary division of the sample, the DM statistics indicate that the predictability is strongest in the 2000-2004 period.

In contrast to the UK results, the DM statistics that correspond to the S&P 500 index suggest that the US dividend ratios fail to outperform the historical moving average confirming our prior belief that the reported differences in Table 3 are not statistically different. Specifically, the computed DM statistics in this case vary between -0.09 and 1.02.⁵

To investigate the dynamic nature of stock return predictability in more detail we turn to the graphical procedure.

4.3 A Graphical Evaluation Method for the Out-of-Sample Performance

Figure 2 depicts the relevant graph of the recursive residuals diagnostic test described in section 3.3 when applied to the FTSE All Share index.

[Insert Fig. 2 around here]

The graph shows that, relative to the historical mean, both the dividend-price and the dividend yield do not exhibit superior performance during the 1993-1994 period. Additionally, towards the end of 1998 and until mid-1999 the slope is strongly negative indicating a superior performance of the historical mean. However, during 1994-1996 and particularly post-1999, the graph line reveals an upward tendency (i.e. the slope is positive) which prevails until the end of the sample, in 2004, suggesting a better performance of the dividend models. The usefulness of the graphical diagnostic in this context is clear: it confirms the predictability results from the conventional

⁵ Using asymptotic critical values from McCracken (2007) does not affect our results. In this case, the UK dividend ratios would significantly outperform the prevailing mean also at the 1% level during the full out-of-sample period. Additionally, the UK dividend yield and the US dividend-price ratio would significantly outperform the historical mean at the 5% during the 1990-1995 period.

measures and clarifies that the period from the late 1990s onwards is highly predictable.

To facilitate a more direct comparison, Figure 3 shows the graph of the recursive residuals diagnostic test for the S&P 500 index as plotted against the corresponding graph for the FTSE All Share index that was depicted in figure 2.⁶

[Insert Fig. 3 around here]

Again, the graphical diagnostic confirms the results from the more conventional methods and indicates that the US dividend models fail to outperform the historical moving average model out-of-sample. In particular, the graph line crosses the zero line quite frequently and it is consistently below the graph line that is associated with the UK market. Although we observe a positive slope during the periods 1994-1995 and 1999-2001, the graph line starts from a negative area in both cases. In 2001 onwards, the gap between the two lines becomes remarkably wide indicating that dividend ratios are more informative in the UK during that period.

In the next section we offer a detailed discussion and provide empirical evidence with the purpose of giving a thorough interpretation to our findings.

4.4 Explanation of Predictability Findings and Further Empirical Evidence

4.4.1 Discussion

As noted above, this study offers a possible explanation of why predictability results in the UK differ to those reported in recent US studies. We believe that the study of Fama and French (2001) can help us explain the poor performance of the dividend ratios when used to predict US stock returns and reconcile the contradictory findings.

⁶ The graph in Figure 3 is obtained when the dividend yield is the predictive variable. In any case, the figure would be analogous if the dividend-price ratio was employed instead.

In more detail, Fama and French (2001) present the phenomenon of the disappearing dividend in the US context. Their study documents the tendency of public companies to pay less dividends since 1978. In particular, from a post-1972 peak of 66.5% in 1978, the percent of dividend payers among NYSE, AMEX and NASDAQ non-financial non-utility firms falls to 20.8% in 1999. The evidence suggests that three fundamentals mainly affect the decision to pay dividends: profitability, investment opportunities and size. Dividend payers tend to be large profitable firms and firms that have never paid are smaller, less profitable but also with more investment opportunities (e.g. higher asset growth rates, higher ratio of the market value of assets to their book value).

The source of a lower propensity to pay is the changing characteristics of newly listed firms (after 1978) which differ from earlier new lists. New listed companies have low profitability and strong investment opportunities; the well-known characteristics of firms that have never paid dividends.⁷

In a more recent study, Amihud and Li (2006) also propose an explanation for the “disappearing dividend” phenomenon in the US market, which they argue is partly due to a decline in the information content of dividend announcements. Thus, if dividends have become less informative and considering that they entail costs, firms may want to avoid initiating them or may even reduce them. An also important finding which can partly explain the disappearing dividends is the documented increase in institutional holdings. It is suggested that institutional investors are more sophisticated and informed than average individual investors and by the time dividend news is announced, the information that it is intended to convey is already

⁷ Fama and French (2001) find that, even after controlling for characteristics, firms become less likely to pay dividends. This result implies that the perceived benefits of dividends decline through time.

incorporated in the stock price. In addition, firms with high institutional holdings seem less likely to raise their dividends.

In the UK, little work has addressed the issue. Nevertheless, recent evidence by Benito and Young (2003) provides an interesting comparison with the US studies. They identify an increase in the proportion of quoted UK companies that omit dividend payments from 1995. More specifically, the proportion of dividend-issuing companies stood at 94.6% in 1979 but decreased to 83.9% and 82.1% in 1982 and 1992 respectively, both periods of recession. The proportion had decreased further by 1999 with 74.8% of quoted UK companies issuing dividends. These findings indicate a decline in the propensity of UK companies to pay dividends which however, is not of the same magnitude as in the US market.

4.4.2 Further Empirical Evidence on Dividends: Data and Results

In this section we provide evidence on the annual proportions of quoted UK and US companies that issue dividends throughout the period of our study. The data used were obtained from company accounts held on Datastream and cover the period from 1974 to 2004. As in Benito and Young (2003), we use Datastream's lists of all live and dead UK companies.⁸ The US data cover firms that are listed on AMEX, NASDAQ and NYSE indices.⁹ We also provide a direct comparison between companies listed on the FTSE All Share and the S&P 500 indices (i.e. the focus of our empirical application).

[Insert Fig. 4 around here]

Figure 4 i) shows the proportions of all quoted UK and US companies that actually issue dividends during the period 1974-2004. We observe a long decreasing

⁸ Dead companies include those that subsequently failed, merged or de-listed.

⁹ In contrast to Fama and French (2001) who use only publicly traded non-financial non utility firms, we employ all quoted US companies to produce results that are more comparable to the UK case.

trend for both markets suggesting that less companies pay dividends through time. However, there is a consistent difference between the two markets suggesting that a culture of dividends is much more associated with the UK market rather than the US. In 1979, 89.2% of UK companies issued dividends while the corresponding percentage for US companies was 76.7%. This difference remarkably widened in 1999 when the proportion of UK companies paying dividends fell to 75.8% and the proportion of US companies reached a low of 54.5%. This trend seems to start reversing after the year 2003 but the fact remains that there is an important difference between the two markets for the whole out-of-sample period under examination (i.e. 1990-2004). This gives support to our findings and explains why in general, dividends might be more informative in the UK context.

Figure 4 ii) isolates the corresponding proportions of companies listed on the UK FTSE All Share and those listed on the S&P 500 indices which are considered in this study. In 1979, 97.2% of companies listed on the FTSE All Share index and 92.5% of companies listed on the S&P 500 issued dividends. In 1999 however, the corresponding proportions fell to 89.7% and 72.2% respectively, significantly broadening the difference between the two indices. The identified difference provides further support to our previous findings and sheds more light on our results. Even though the S&P 500 index includes large companies which are more likely to pay dividends, there is still a significant divergence and the percentage of companies listed on the FTSE All Share issuing dividend payments is higher. This is particularly evident during the out-of-sample period considered in this study and strengthens the conjecture that dividends might convey more information in the UK rather than the US market during that period.

To complement the above evidence, we further explored whether our theory is indeed in line both with our in-sample and out-of-sample results. First, as Figure 4 ii) suggests, the fraction of dividend-issuing companies in the S&P 500 index starts to fall in 1980 as opposed to 1975 which is the starting point of our sample. Therefore, we estimated in-sample, two appropriate non-overlapping sub-samples containing an equal number of observations starting from 1980. The results are shown in Table 5.

[Insert Table 5 around here]

We then find that during the early sub-sample which covers the period between 1980:01 and 1992:05, both dividend ratios are significant in both markets. However, in the second sub-sample which spans 1992:06-2004:10, statistical significance is maintained only in the UK market, even though its has somewhat decreased in magnitude. With respect to the S&P 500, both ratios become insignificant. This finding supports both our theory of decreasing dividends and our out-of-sample results. Furthermore, it can also be seen as another confirmation that our out-of-sample results reflect our in-sample findings.

4.5 The Instrumented Model

GW indicate that it might be possible to enhance the dividend-ratio forecasting coefficients by instrumenting the model to account for the time-varying properties of the dividend-price ratio and dividend growth processes. To our knowledge this has not been applied in a UK context. To that end we apply the methodology outlined in section 3.4 to the FTSE All Share index. Initially, Figure 5 plots the time series of the naïve stock return betas and the instrumented Campbell-Shiller (CS) based betas.¹⁰

[Insert Fig. 5 around here]

¹⁰ For the estimation of the CS betas in equation (11), ρ can be calibrated to be about 0.96 for the UK market.

The CS betas show a decline during the decade 1990-2000 and are typically slightly lower than the ordinary betas. However the trend reverses from 2000 and reveals an upward tendency while at the same time the CS betas are slightly higher than the ordinary betas.

Table 6 shows how the prevailing mean, the straight dividend model and the instrumented dividend model perform when predicting stock returns. The forecast error statistics are calculated with respect to the period from 1990:01 to 2004:10 (i.e. our out-of-sample period). Note that direct forecasts in Table 6 are comparable to those in Table 3 with the only difference being that the former forecasts stock returns themselves while the latter is constructing forecasts of excess stock returns.

[Insert Table 6 around here]

The results indicate that both models manage to outperform the prevailing stock return moving average. The RMSE for the prevailing mean model is found to be 4.59% as opposed to 4.03% for the straight dividend model and 3.99% for the instrumented model. In addition, the DM statistics for the straight dividend model and the instrumented model are 3.03 and 3.25 respectively, indicating superior performance against the historical moving average model, both at 1% and 5% levels of significance.

Furthermore, when we compare the straight dividend model against the Campbell and Shiller based model (i.e. the instrumented model) we find that they are statistically different from one another with the latter exhibiting better performance (the DM statistic is 4.09). The last result establishes the fact that we can improve predictive ability using the Campbell and Shiller (1988b) accounting identities. In other words, taking into consideration the time-varying properties of the dividend yield and dividend growth processes by instrumenting our model, we can enhance

forecasting coefficients and improve out-of-sample performance. Again, this new result for the UK is in sharp contrast with Goyal and Welch (2003) and the US context.

5. Conclusion

A relatively small body of literature suggests that, in contrast to recent US studies, a degree of predictability exists in UK equity premia. The current paper extends this work and offers a comparative study between the two markets by primarily employing a recursive residuals (out-of-sample) graphical approach. This methodology allows for a more dynamic identification of predictability, revealing the actual time periods where dividend ratios succeed (or fail) in predicting the equity premium. Its usefulness was first illustrated by Goyal and Welch (2003). They show graphically that the supposed predictive ability of post-war US dividend ratios is essentially dependent on a few outlier observations.

Our empirical application relies on the well known UK FTSE All Share stock market index and uses monthly data covering the 1975:04-2004:10 period. Corresponding data for the US market were derived from the S&P 500 index. Firstly, preliminary in-sample univariate regressions show that the UK dividend ratios have significant forecasting power. The dividend ratios retain in-sample statistical significance also in the US context. In this case however, they produce lower t-statistics (both normal and adjusted) and lower R-squareds compared to the corresponding UK regressions when the full sample is employed. Secondly, as far as the conventional out-of-sample performance is concerned, forecast error statistics and Diebold-Mariano (1995) tests indicate that UK dividend ratio models significantly outperform the unconditional historical equity premium model (the prevailing simple

moving average). Nevertheless, the dividend models do not manage to outperform the historical moving average model in the US market at the usual levels of significance. These preliminary results can be seen as further confirmation of the literature that suggests dividend ratios can be employed to predict UK equity premia.

To analyse the dynamic nature of equity premia predictability we next turn to the graphical procedure. The general idea is to plot (against time) the cumulative sum-squared error from an unconditional model minus the cumulative sum squared error from a dividend ratio model. An upward tendency in the graph line therefore suggests better performance of the dividend ratio models. GW stress that although a simple procedure, its neglect by the literature may have left useful information regarding predictability uncovered. When the methodology is applied to our UK data, pockets of forecastability are shown to exist before the late 1990s. Most strikingly however, the methodology identifies the period from the late 1990s onwards as consistently predictable. This new result adds significantly to the information set regarding UK equity premia predictability: for example, revealing that unlike in the US case, predictability is not the result of outlier observations.

Employing the graphical method with US data, we do not find evidence of out-of-sample predictability. The graph line often crosses the zero line and it is constantly below the corresponding line derived from the FTSE All Share index. This is particularly evident from 2001 onwards with the occurrence of a significant spread between the two markets.

Our paper provides a possible explanation of why a predictable component is identified in the UK and not in the US stock returns. Our theory is linked to the study of Fama and French (2001) who first present the phenomenon of the disappearing dividend in the US market. More recently, Amihud and Li (2006) extend their work

and also propose an explanation for the “disappearing dividend”. Extremely limited work around this issue has been documented with respect to the UK market (e.g., Benito and Young, 2003). Taking into consideration the above studies we obtained annual dividends data for both markets and we report the proportions of companies that actually issue dividends during the 1974-2004 period. Our results confirm the existing evidence which reveals the propensity of companies to pay less dividends but we also come across an important difference between the UK and US companies. In fact, throughout the whole sample period and especially during the out-of-sample period considered in this paper, the percentage of UK dividend-issuing companies is much higher than the corresponding percentage for US companies. Therefore, we argue that dividend ratios are perhaps more informative in the UK context and can explain a larger fraction of the UK stock market returns, as shown in this study.

Finally, in an effort to improve the predictability of UK equity premia, we employed Campbell and Shiller’s (1988b) theory to instrument the forecasting model. Specifically, we account for the time-varying properties of the dividend ratio and dividend growth processes. To our knowledge this has not been applied in a UK context. In contrast to GW we find that the relevant forecast errors and the Diebold and Mariano (1995) statistics reveal that the Campbell and Shiller (1988b) identities can further improve the forecasting ability of UK dividend ratios.

References

- Amihud, Y., Li, K., 2006. The declining information content of dividend announcements and the effects of institutional holdings. *Journal of Financial and Quantitative Analysis* 41(3), 637–660.
- Ang, A., Bekaert, G., 2007. Stock return predictability: is it there? *Review of Financial Studies* 20(3), 651–707.
- Benito, A., Young, G., 2003. Hard times or great expectations? Dividend omissions and dividend cuts by UK firms. *Oxford Bulletin of Economics and Statistics* 65(5), 531–555.
- Bera, A.K., Jarque, C.M., 1980. Efficient tests for normality, homoskedasticity, and serial independence of regression residuals. *Economic Letters* 6(3), 255–259.
- Bossaerts, P., Hillion, P., 1999. Implementing statistical criteria to select return forecasting models: what do we learn? *Review of Financial Studies* 12(2), 405–428.
- Campbell, J.Y., 1987. Stock returns and the term structure. *Journal of Financial Economics* 18(2), 373–399.
- Campbell, J.Y., Shiller, R.J., 1988a. Stock prices, earnings, and expected dividends. *Journal of Finance* 43(3), 661–676.
- Campbell, J.Y., Shiller, R.J., 1988b. The dividend-price ratio and expectations of future dividends and discount factors. *Review of Financial Studies* 1(3), 195–228.
- Campbell, J.Y., Shiller, R.J., 1998. Valuation ratios and the long-run stock market outlook. *Journal of Portfolio Management* 24(2), 11–26.

- Dickey, D.A., Fuller, W.A 1979. Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association* 74(366), 427–431.
- Diebold, F.X., Mariano, R.S., 1995. Comparing predictive accuracy. *Journal of Business and Economic Statistics* 13(3), 253–263.
- Fama, E.F., French, K.R., 1988. Dividend yields and expected stock returns. *Journal of Financial Economics* 22(1), 3–25.
- Fama, E.F., French, K.R., 1989. Business conditions and expected returns on stocks and bonds. *Journal of Financial Economics* 25(1), 23–49.
- Fama, E.F., French, K.R., 2001. Disappearing dividends: changing firm characteristics or lower propensity to pay? *Journal of Financial Economics* 60, 3–44.
- Goyal, A., Welch, I., 2003. Predicting the equity premium with dividend ratios. *Management Science* 49(5), 639–654.
- Hodrick, R.J., 1992. Dividend yields and expected stock returns: alternative procedures for inference and measurement. *Review of Financial Studies* 5(3), 357–386.
- Inoue, A., Killian, L., 2004. In-sample or out-of-sample tests of predictability: which one should we use? *Econometric Reviews* 23(4), 371–402.
- Keim, D.B., Stambaugh, R.F., 1986. Predicting returns in the stock and bond markets. *Journal of Financial Economics* 17(2), 357–390.
- Kothari, S.P., Shanken, J., 1997. Book-to-market, dividend yield, and expected market returns: a time series analysis. *Journal of Financial Economics* 44(2), 169–203.

- Lamont, O., 1998. Earnings and expected returns. *Journal of Finance* 53(5), 1563–1587.
- Lettau, M., Ludvigson, S., 2001. Consumption, aggregate wealth and expected stock returns. *Journal of Finance* 56(3), 815–849.
- Lewellen, J., 2004. Predicting returns with financial ratios. *Journal of Financial Economics* 74(2), 209–235.
- McCracken, M.W., 2007. Asymptotics for out-of-sample tests of Granger causality. *Journal of Econometrics* 140(2), 719–752.
- McMillan, D.G., 2003. Non-linear predictability of UK stock market returns. *Oxford Bulletin of Economics and Statistics* 65(5), 557–573.
- Newey, W., West, K.D., 1987. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica* 55(3), 703–708.
- Pesaran, M.H., Timmermann, A., 2000. A recursive modelling approach to predicting UK stock returns. *The Economic Journal* 110(460), 159–191.
- Pontiff, J., Schall, L.D., 1998. Book-to-market ratios as predictors of market returns. *Journal of Financial Economics* 49(2), 141–160.
- Rapach, D.E., Wohar, M.E., 2006. In-sample vs. out-of-sample tests of stock return predictability in the context of data mining. *Journal of Empirical Finance* 13(2), 231–247.
- Rozeff, M., 1984. Dividend yields are equity risk premiums. *Journal of Portfolio Management* 11, 68–75.

APPENDIX

1 TABLES

Table 1 Descriptive Statistics

	Mean	Sdev.	Median	Min	Max	Skew	Kurt	Br-Jq	ADF
Sample 1975:04-2004:10									
FTSE ALL SHARE									
$r_{m,t}$	5.08	5.25	5.24	-33.95	29.12	-0.86	12.25	1311.44***	-17.92***
EQP_t	5.29	9.16	5.55	-29.48	34.39	-0.24	4.59	41.17***	-16.26***
DP_t	-3.20	0.31	-3.17	-3.97	-2.53	-0.42	2.60	13.05***	-1.99
DY_t	-3.19	0.30	-3.15	-3.90	-2.58	-0.51	2.51	19.09***	-1.83
Δd_t	0.68	7.66	0.98	-37.5	46.43	0.12	8.51	449.38***	-14.46***
S&P 500									
$r_{m,t}$	3.95	4.42	4.10	-19.75	16.05	-0.75	5.76	146.02***	-17.09***
EQP_t	4.25	8.47	4.02	-17.76	39.39	0.76	5.13	100.97***	-15.93***
DP_t	-3.53	4.87	-3.43	-4.59	-2.66	-0.44	2.07	24.48***	-0.67
DY_t	-3.51	4.84	-3.41	-4.53	-2.75	-0.47	2.05	26.43***	-0.74
Δd_t	0.48	5.97	0.58	-28.50	30.58	-0.17	5.87	123.98***	-17.19***

Explanation: All series are described in section 2. $r_{m,t}$ is the log return on the UK FTSE All Share and the S&P 500 stock market indices from month $t-1$ to t . The equity premium EQP_t subtracts the equivalent log return on a 3-month treasury bill. $DP(t)$ is the dividend-price ratio, i.e. the log of the dividend D_t divided by the stock market price P_t . The dividend yield DY_t divides by P_{t-1} instead. Δd_t is the change in log dividends from month $t-1$ to t . All variables are in percentages. Br-Jq is the Bera and Jarque (1980) test for normality. The critical level to reject normality is 9.21 at the 99% level, 5.99 at the 95% level and 4.61 at the 90% level. ADF is the augmented Dickey-Fuller (Dickey and Fuller 1979) test for the presence of a unit root. An ADF value of -3.5 rejects the presence of a unit root at the 1% level (-2.9 at the 5% level; -2.6 at the 10% level). Three stars indicate significance at the 1% level.

Table 2 In-Sample Univariate Regressions $EQP_t = \alpha + \beta x_{t-1} + \varepsilon_t$

x_{t-1}	α	β	$R^2\%$	$R^2\text{adj.}\%$	s.e.%	α	β	$R^2\%$	$R^2\text{adj.}\%$	s.e.%
Full Sample 1975:04-2004:10	FTSE ALL SHARE					S&P 500				
DP_{t-1}	0.236	0.058	3.85	3.57	8.88	0.135	0.026	2.27	2.00	8.38
t-stat	[4.80]	[3.76]				[4.14]	[2.87]			
NW t-stat	[4.11]	[3.32]				[3.26]	[2.27]			
DY_{t-1}	0.253	0.063	4.49	4.22	8.85	0.131	0.025	2.05	1.77	8.39
t-stat	[5.11]	[4.08]				[3.99]	[2.72]			
NW t-stat	[4.36]	[3.60]				[3.13]	[2.14]			
Sample Period 1975:04-1995:04										
DP_{t-1}	0.290	0.076	2.16	1.75	10.09	0.246	0.062	3.04	2.64	8.54
t-stat	[2.89]	[2.30]				[3.38]	[2.74]			
NW t-stat	[2.37]	[1.90]				[2.50]	[2.08]			
DY_{t-1}	0.370	0.103	3.40	2.99	10.02	0.243	0.061	2.78	2.37	8.56
t-stat	[3.45]	[2.90]				[3.23]	[2.61]			
NW t-stat	[2.89]	[2.46]				[2.42]	[2.00]			
Sample Period 1975:04-2000:04										
DP_{t-1}	0.228	0.055	2.57	2.24	9.36	0.153	0.032	2.72	2.39	8.08
t-stat	[3.72]	[2.81]				[4.04]	[2.89]			
NW t-stat	[3.15]	[2.43]				[3.57]	[2.71]			
DY_{t-1}	0.254	0.063	3.22	2.90	9.33	0.149	0.031	2.45	2.12	8.09
t-stat	[4.04]	[3.16]				[3.88]	[2.74]			
NW t-stat	[3.40]	[2.71]				[3.46]	[2.59]			

Explanation: Table 2 presents the results of the following univariate regression:

$$EQP_t = \alpha + \beta x_{t-1} + \varepsilon_t$$

For each regression the estimated coefficients are given in the first row while the OLS t-statistics and the Newey-West adjusted t-statistics are given in the second and third row respectively. Results are reported with respect to the FTSE All Share and the S&P 500 indices.

Interpretation: The dividend ratios are significant in both markets. However, when the full sample is used and we have maximum degrees of freedom, the UK regressions produce higher t-statistics (both normal and adjusted) and higher R-squareds compared to the corresponding US regressions.

Table 3 Out-of-Sample Performance: Forecast Errors

Forecast Error Statistic	Prevailing Mean %	Dividend- Price Ratio Model %	Dividend Yield Model %	Prevailing Mean %	Dividend- Price Ratio Model %	Dividend Yield Model %
Full Sample 1990:01-2004:10	FTSE ALL SHARE			S&P 500		
Mean	-1.45	0.49	0.96	-1.13	1.04	0.98
Standard Deviation	6.51	6.35	6.31	7.66	7.59	7.61
Root Mean Square Error	6.65	6.36	6.36	7.72	7.64	7.65
Mean Absolute Error	5.15	4.81	4.79	5.84	5.74	5.74
Subsample 1990:01-1995:01						
Mean	0.03	0.91	1.27	-1.07	0.64	0.62
Standard Deviation	7.85	7.72	7.55	7.18	7.01	7.04
Root Mean Square Error	7.79	7.71	7.60	7.20	6.99	7.00
Mean Absolute Error	6.09	6.00	5.79	5.57	5.36	5.38
Subsample 1995:02-2000:02						
Mean	-1.66	1.03	1.73	-1.11	1.77	1.64
Standard Deviation	5.67	5.61	5.63	5.72	5.59	5.62
Root Mean Square Error	5.86	5.66	5.85	5.78	5.82	5.82
Mean Absolute Error	4.33	4.25	4.37	4.42	4.67	4.66
Subsample 2000:03-2004:10						
Mean	-2.80	-0.57	-0.21	-1.25	0.70	0.70
Standard Deviation	5.45	5.37	5.38	9.83	9.85	9.85
Root Mean Square Error	6.10	5.35	5.34	9.82	9.78	9.79
Mean Absolute Error	5.00	4.13	4.14	7.68	7.30	7.30

Explanation: All series are described in section 2 and Table 1. This table presents the properties of the equity premium prediction errors from the naïve model that uses only the prevailing historical mean equity premium as a forecast and another model that uses the dividend-price ratio or the dividend yield as a predictor. Both models use all prevailing data beginning in 1975:04. Bold-face values denote superior performance.

Table 4 Diebold and Mariano (1995) Statistics

Sample Period	Dividend-Price Ratio Model	Dividend Yield Model	Dividend-Price Ratio Model	Dividend Yield Model
	FTSE ALL SHARE		S&P 500	
1990:01-2004:10 (full sample)	2.13 ^{**}	1.70 ^{**}	0.48	0.43
1990:01-1995:01	0.80	1.40	1.02	0.92
1995:02-2000:02	0.57	0.07	-0.09	-0.09
2000:03-2004:10	2.75 ^{***}	2.47 ^{***}	0.13	0.12

Explanation: This table tabulates the computed Diebold and Mariano (1995) statistics across different out-of-sample periods. The Diebold and Mariano (1995) statistic tests whether the RMSE performances

reported in Table 3 are statistically different from one another. Two stars indicate significance at the 5% level and three stars indicate significance at the 1% level.

Interpretation: For the full out-of-sample period, the UK dividend models significantly outperform the historical moving average at the 5% significance level. During the last period of our sample, the dividend ratios perform better than the unconditional mean both at 5% and 1% levels of significance. However, Table 4 suggests that forecasts derived from the US dividend models are not significantly different from forecasts produced by the unconditional mean model.

Table 5 In-Sample Univariate Regressions (sub-samples) $EQP_t = \alpha + \beta x_{t-1} + \varepsilon_t$

x_{t-1}	α	β	$R^2\%$	$R^2 \text{ adj.}\%$	α	β	$R^2\%$	$R^2 \text{ adj.}\%$
Sample Period 1980:01-1992:05	FTSE ALL SHARE				S&P 500			
DP_{t-1}	0.471	0.134	7.60	6.97	0.317	0.081	4.01	3.36
t-stat	3.99	3.48			3.02	2.48		
NW t-stat	3.31	2.83			2.16	1.82		
p-value (normal t-stat)	[0.000]	[0.001]			[0.003]	[0.014]		
p-value (NW t-stat)	[0.001]	[0.005]			[0.033]	[0.072]		
DY_{t-1}	0.485	0.139	6.85	6.22	0.324	0.083	3.87	3.22
t-stat	3.76	3.29			2.95	2.43		
NW t-stat	3.27	2.81			2.16	1.83		
p-value (normal t-stat)	[0.000]	[0.001]			[0.004]	[0.016]		
p-value (NW t-stat)	[0.001]	[0.006]			[0.032]	[0.070]		
Sample Period 1992:06-2004:10								
DP_{t-1}	0.256	0.062	5.57	4.93	0.082	0.013	0.28	-0.40
t-stat	3.49	2.95			1.01	0.64		
NW t-stat	2.55	2.18			0.98	0.62		
p-value (normal t-stat)	[0.001]	[0.004]			[0.313]	[0.522]		
p-value (NW t-stat)	[0.012]	[0.031]			[0.327]	[0.535]		
DY_{t-1}	0.282	0.070	6.88	6.25	0.066	0.009	0.13	-0.55
t-stat	3.84	3.30			0.81	0.44		
NW t-stat	2.70	2.35			0.78	0.42		
p-value (normal t-stat)	[0.000]	[0.001]			[0.418]	[0.660]		
p-value (NW t-stat)	[0.008]	[0.020]			[0.439]	[0.676]		

Explanation: Table 5 presents the results of the following univariate regression in two different sub-sample periods:

$$EQP_t = \alpha + \beta x_{t-1} + \varepsilon_t$$

For each regression the estimated coefficients are given in the first row while the OLS t-statistics and the Newey-West adjusted t-statistics are given in the second and third row respectively. Finally, the corresponding p-values are given in the fourth and fifth row. Results are reported with respect to the FTSE All Share and the S&P 500 indices.

Interpretation: In the early sub-sample period, the dividend ratios are significant in both markets. However, significance is maintained only in the UK market when the late period is considered, even though it has somewhat decreased in magnitude. This finding is consistent both with our theory of decreasing dividends and our out-of-sample results.

Table 6 Instrumented Dividend-Ratio Forecasts for Returns

Forecast Error Statistic	Prevailing Mean %	Straight Dividend Model %	Instrumented Dividend Model %
Mean	-1.90	0.20	0.17
Standard Deviation	4.19	4.04	4.00
Root Mean Square Error	4.59	4.03	3.99
Mean Absolute Error	3.44	3.12	3.09

Explanation: This table reports the UK results of stock return prediction from a model that uses the prevailing average stock return as a forecast and two other models: The straight dividend model which produces forecasts using OLS and the instrumented dividend model which uses the Campbell-Shiller betas to construct forecasts (see section 3.4 and equations (11) – (16) for a detailed description).

The recursive betas are calculated using the entire history of data available. The forecast error statistics involve only the period from 1990:01 to 2004:10. Bold-face values denote superior performance. Note that direct forecasts in Table 6 are comparable to those in Table 3, with the only difference that Table 6 forecasts stock returns instead of excess stock returns. For this period we also calculated the Diebold and Mariano (1995) statistic which measures the statistical difference between RMSEs from two models and is asymptotically normally distributed.

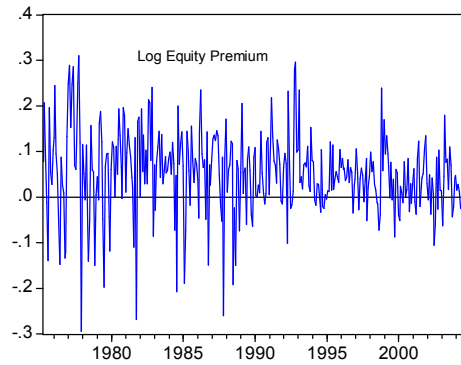
Interpretation: The forecasting ability indeed does improve using the Campbell–Shiller identities. The Diebold and Mariano (1995) statistics for the two models are 3.03 and 3.25 indicating that both models significantly outperform the prevailing moving average out-of-sample. When the straight dividend model is compared to the instrumented model, the Diebold and Mariano (1995) statistic is found to be 4.09 and this further demonstrates that the Campbell–Shiller based betas manage to improve predictive ability.

2 FIGURES

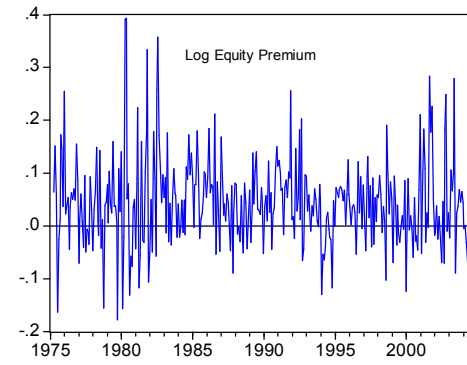
Figure 1 Time Series Graphs

Explanation: The following graphs plot the time series of the log equity premium, the log dividend-price ratio and the log dividend yield respectively. All variables are described in section 2 and in Table 1.

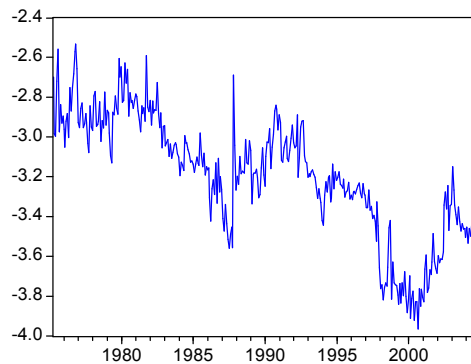
i) The UK Equity Premium



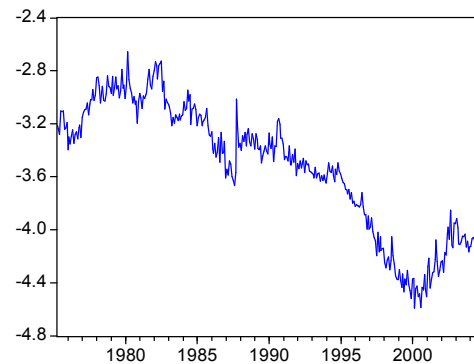
iv) The US Equity Premium



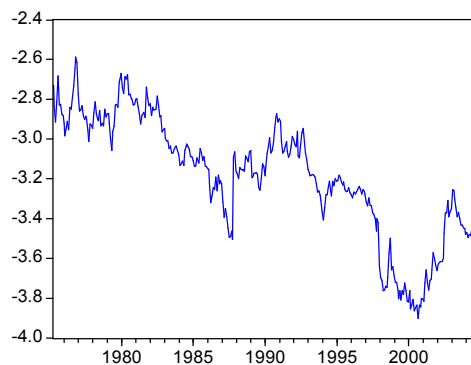
ii) The UK Log Dividend-Price Ratio



v) The US Log Dividend-Price Ratio



iii) The UK Log Dividend Yield



vi) The US Log Dividend Yield

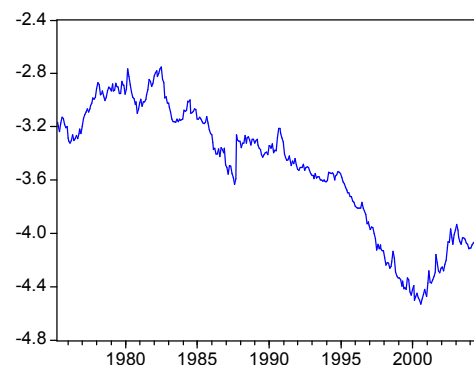
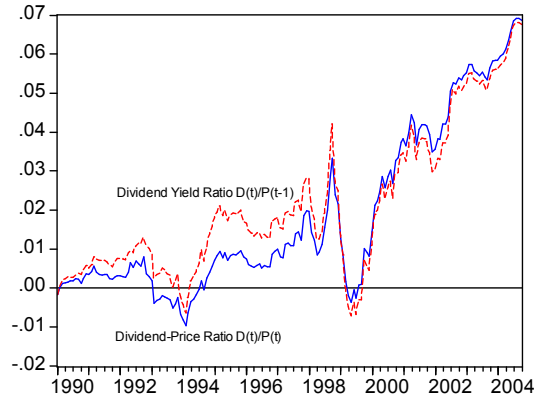


Figure 2 Cumulative Relative Out-of-Sample, Sum-Squared Error Performance: FTSE All Share

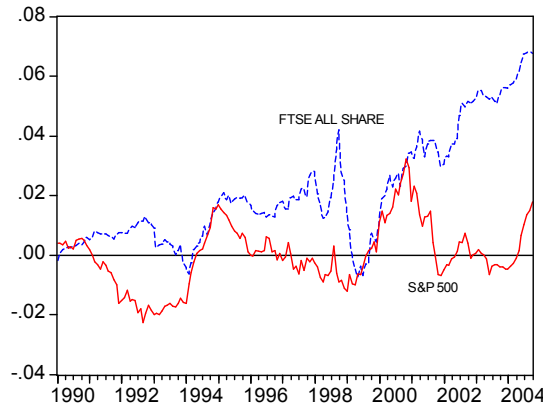


Explanation: Figure 2 plots $Net-SSE_T$ which is the cumulative sum-squared error from the unconditional model minus the cumulative sum-squared error from the dividend ratio model. 1990:01 – 2004:10 is the out-of sample period. In particular, it plots:

$$Net-SSE_T = \sum_{t=1990:01}^T [SE_t^{Prevailing\ Mean} - SE_t^{Dividend\ Model}]$$

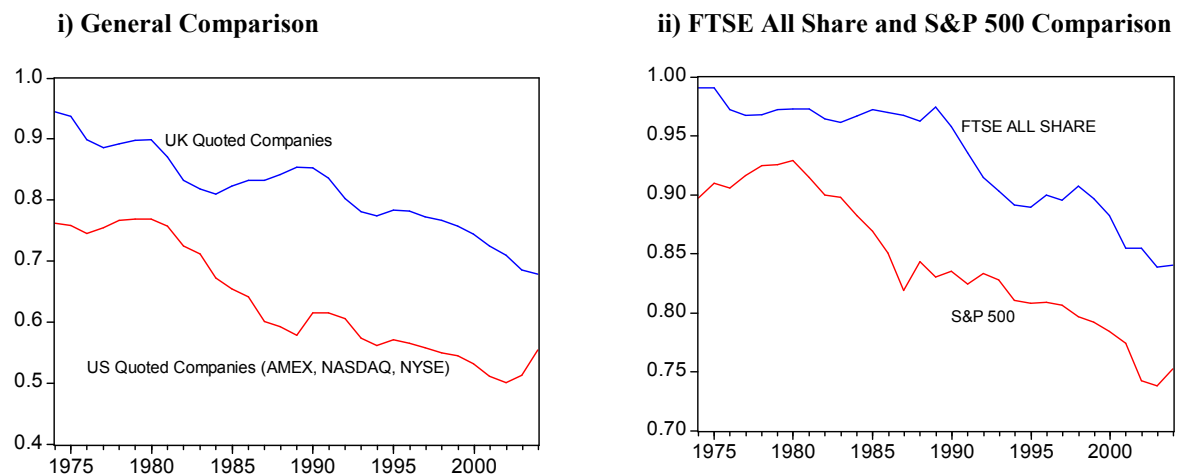
Where SE_t is the squared out-of-sample prediction error in observation t. The unconditional SE is obtained when the prevailing up-to-date equity premium average is used to forecast the following month's equity premium. The conditional prediction errors of the dividend models are obtained from recursive regressions with either DP_{t-1} or DY_{t-1} as the sole predictor of the following month's equity premium.

Figure 3 Cumulative Relative Out-of-Sample, Sum-Squared Error Performance: S&P 500 and FTSE All Share



Explanation: This figure plots $Net-SSE_T$ for the S&P 500 and the FTSE All Share indices (see Section 3.3 and Figure 2 for a detailed description). The graph with respect to FTSE All Share index is taken from Figure 2. Figure 3 depicts the graphical method when the dividend yield is the sole predictor of the following month's equity premium. The out-of-sample period starts from 1990:01 and ends at 2004:10.

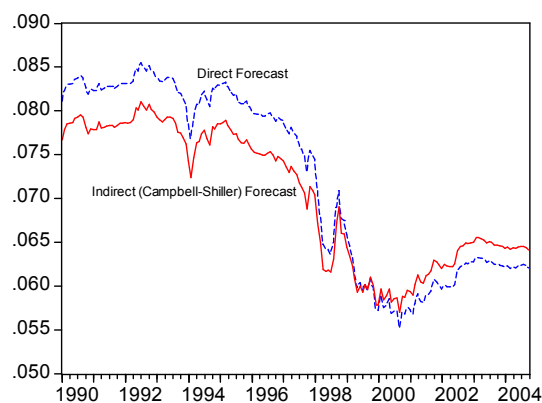
Figure 4 Proportions of Quoted UK and US Companies Issuing Dividends



Explanation: Figure 4 i) plots the annual proportions of quoted UK and US companies that pay cash dividends for the period 1974-2004. UK data includes all dead and active companies and US data cover AMEX, NASDAQ and NYSE indices.

Figure 4 ii) figure plots the annual proportions of companies that pay cash dividends and are listed on the FTSE All Share and S&P 500 indices which are employed for this study. The graph lines span the period from 1974 to 2004.

Figure 5 Campbell-Shiller Betas



Explanation: This figure plots the recursive beta coefficients of forecasts using the dividend-price ratio as a regressor. For a detailed explanation of how direct and indirect (Campbell-Shiller) forecasts are constructed, see section 3.4 and equations (11) – (16).

Note: The recursive betas are calculated using the entire history of data available. Figure 5 plots these betas only for the UK market and for the period 1990:01 to 2004:10.